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(54) **STORAGE LOCATION SELECTION ACCORDING TO QUERY EVALUATION**

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G06F 16/13 (2019.01)
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(52) **U.S. Cl.**
CPC ***G06Q 10/06315*** (2013.01); ***G06F 16/13*** (2019.01); ***G06F 16/24565*** (2019.01); ***G06F 16/24578*** (2019.01); ***G06Q 50/40*** (2024.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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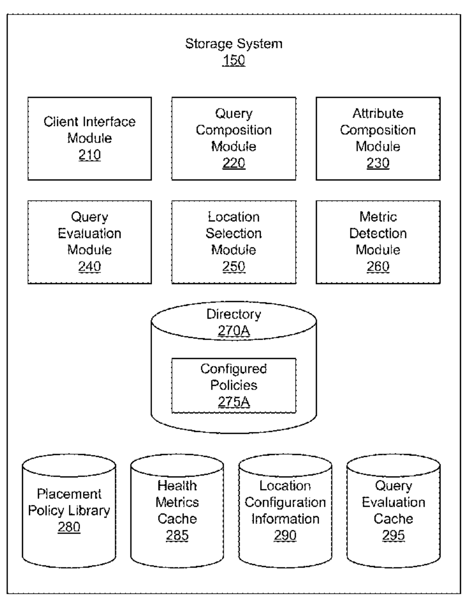
(57) **ABSTRACT**

A method or a system for storing a file from a client device in multiple storage locations based on a placement policy with varied constraints for different copies of the file. The placement policy includes sets of policy constraints for different copies, specifying whether each copy should be uploaded synchronously or asynchronously. For each copy of the file, a specific query is generated based on its associated set of policy constraints. These queries are used to evaluate the attributes of various storage locations, selecting a candidate storage location for each copy. The file copies are then stored at their respective selected locations according to the synchronous or asynchronous upload requirement specified by their respective policy constraints.

20 Claims, 4 Drawing Sheets



(51) **Int. Cl.**

G06F 16/2455	(2019.01)
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2011/0167230 A1 * 7/2011 Quan G06F 3/061 711/170
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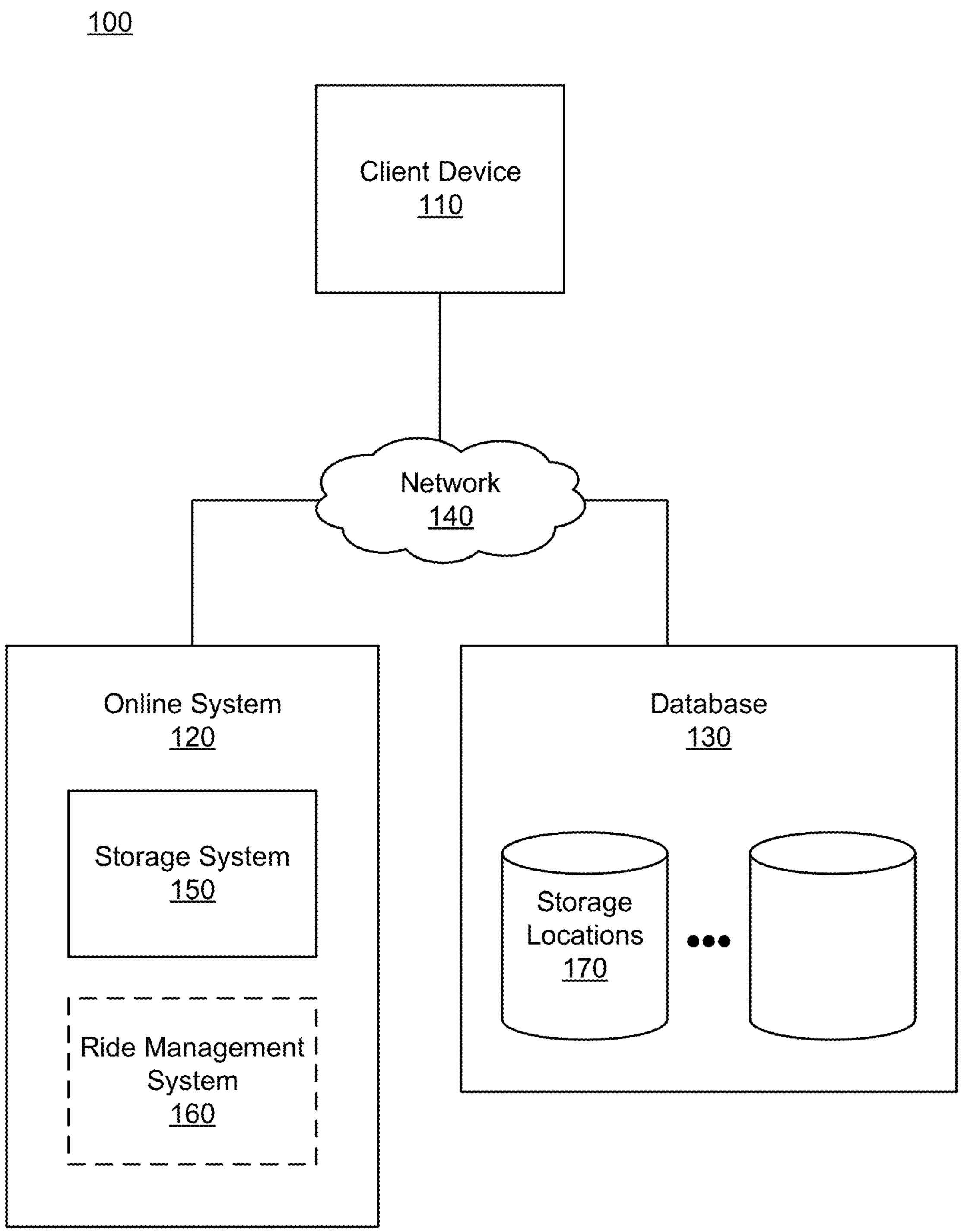
100
Client Device
110
Network
140
Online System
120
Storage System
150
Ride Management
System
160
Database
130
Storage
Locations
170

FIG. 1

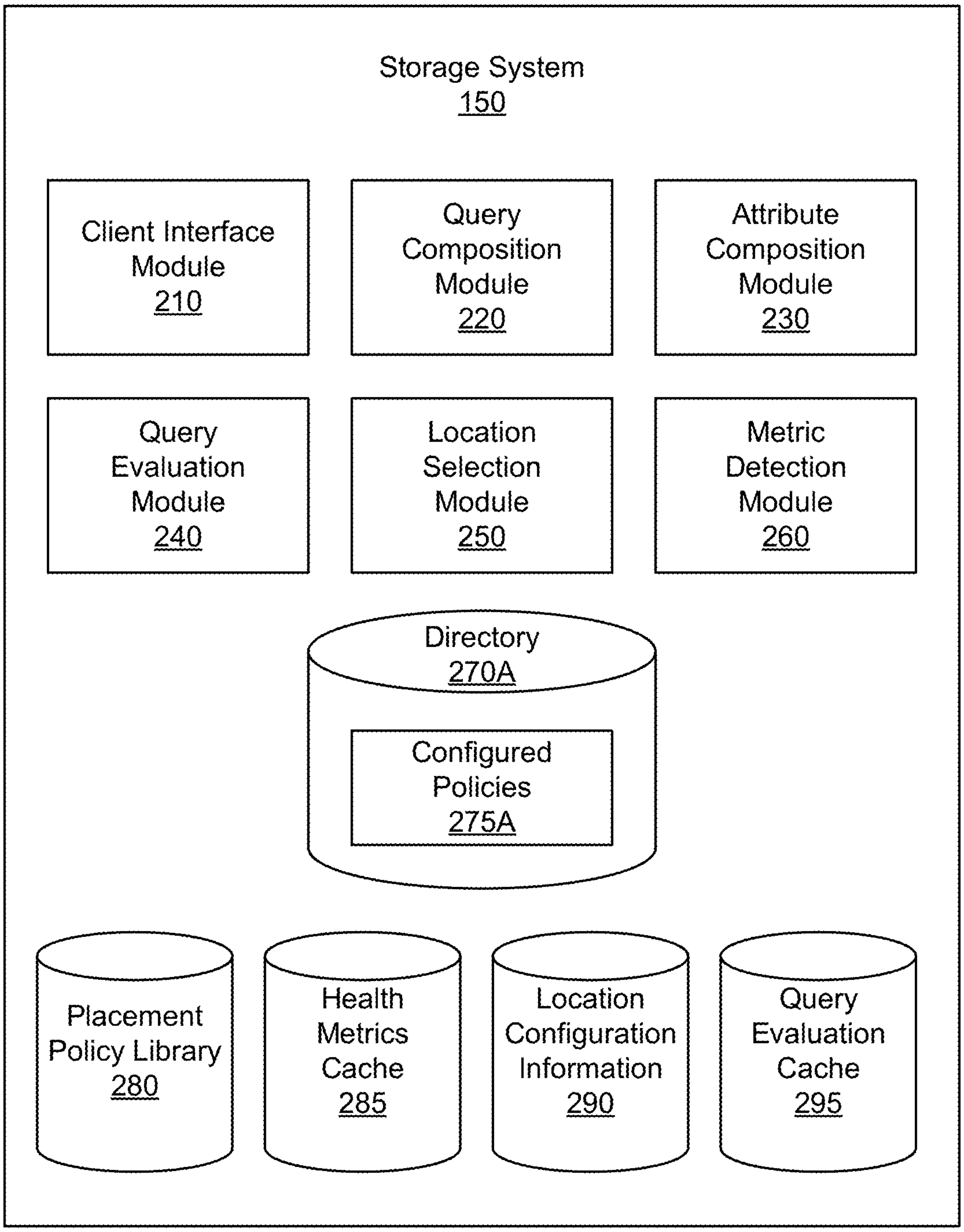
Storage System
150
Client Interface Module
210
Query Composition Module
220
Attribute Composition Module
230
Query Evaluation Module
240
Location Selection Module
250
Metric Detection Module
260
Directory
270A
Configured Policies
275A
Placement Policy Library
280
Health Metrics Cache
285
Location Configuration Information
290
Query Evaluation Cache
295

FIG. 2

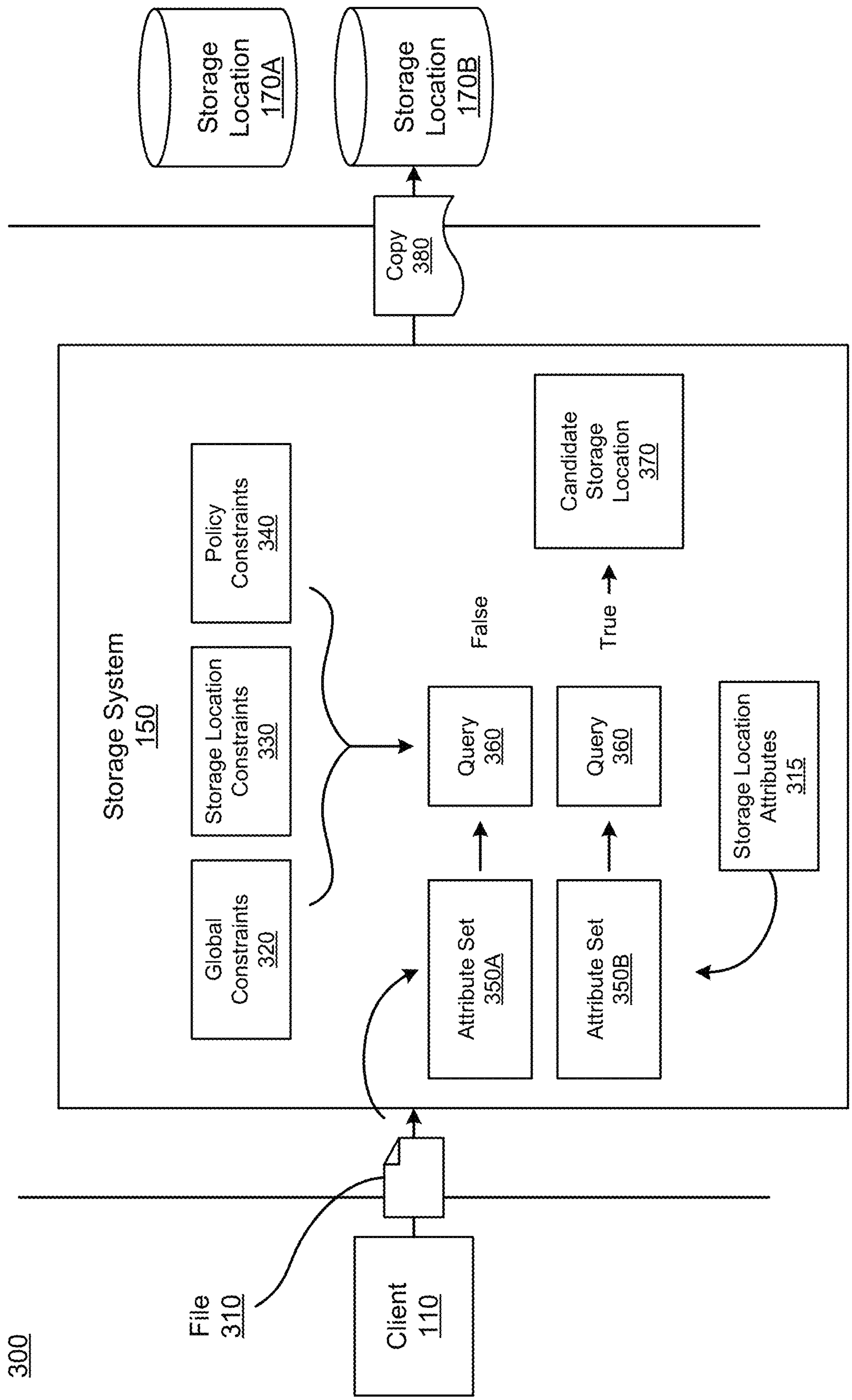
300
Client 110
File 310
Storage System 150
Global Constraints 320
Storage Location Constraints 330
Policy Constraints 340
Attribute Set 350A
Attribute Set 350B
Query 360
Query 360
False
True
Candidate Storage Location 370
Storage Location Attributes 315
Copy 380
Storage Location 170A
Storage Location 170B

FIG. 3

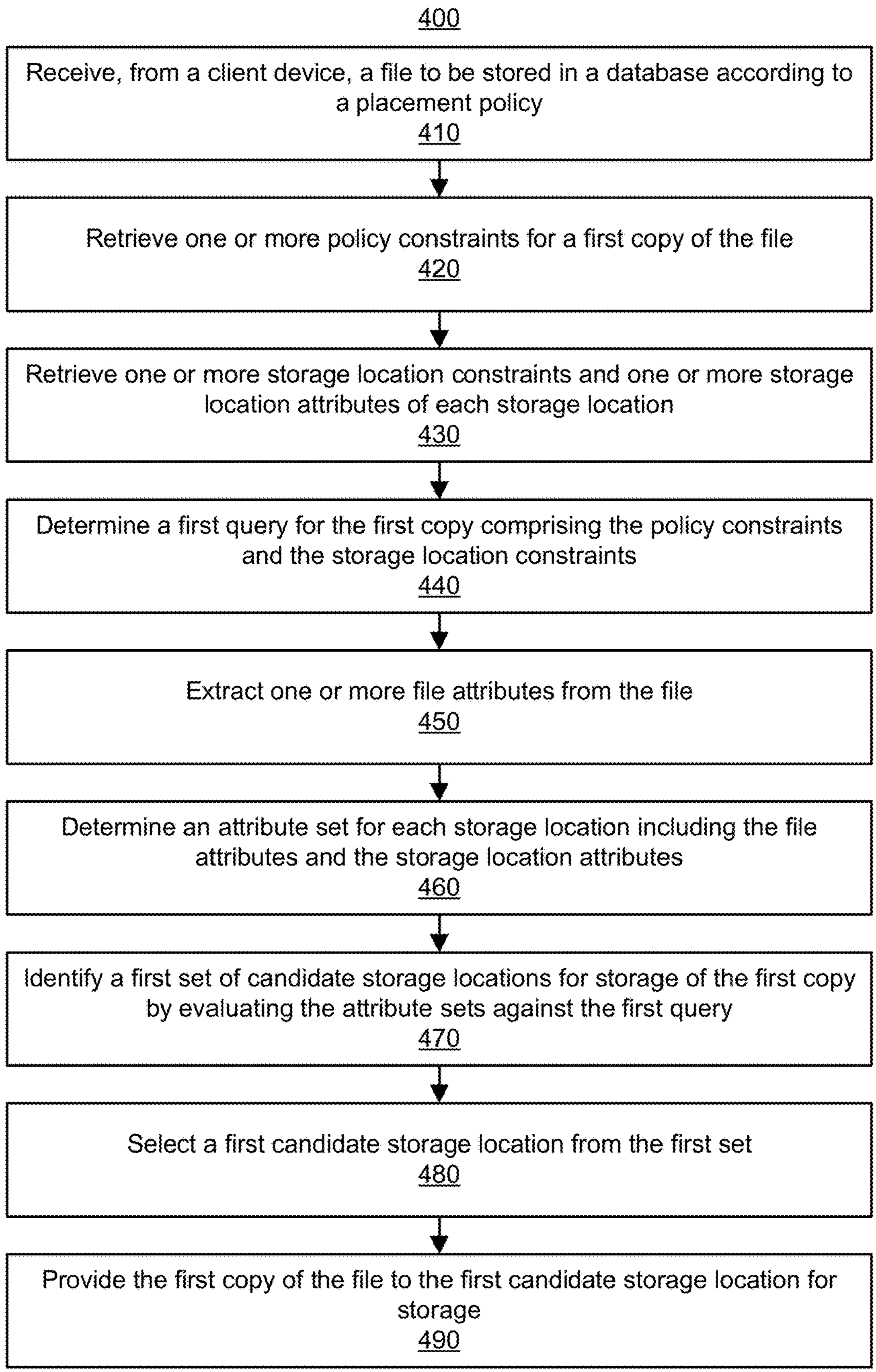
400
Receive, from a client device, a file to be stored in a database according to a placement policy
410
Retrieve one or more policy constraints for a first copy of the file
420
Retrieve one or more storage location constraints and one or more storage location attributes of each storage location
430
Determine a first query for the first copy comprising the policy constraints and the storage location constraints
440
Extract one or more file attributes from the file
450
Determine an attribute set for each storage location including the file attributes and the storage location attributes
460
Identify a first set of candidate storage locations for storage of the first copy by evaluating the attribute sets against the first query
470
Select a first candidate storage location from the first set
480
Provide the first copy of the file to the first candidate storage location for storage
490

FIG. 4

STORAGE LOCATION SELECTION ACCORDING TO QUERY EVALUATION

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 16/805,448, filed Feb. 28, 2020, which is incorporated by reference in its entirety.

BACKGROUND

This present disclosure generally relates to a data storage architecture for computer-based databases and more specifically to selection of one or more storage locations for storage of data.

Traditional computer-based storage systems manage storage and access of mounds of data. These storage systems typically have finite resources and must consider how to efficiently utilize the resources. Placement policies in place for these storage systems traditionally maintain a hierarchy of storage locations. Storage locations are considered according to the hierarchy to identify an available storage location to store a file. However, these placement policies are not easily adaptable to changes in the storage locations, i.e., adding of storage locations, removing of storage locations, changes in the existing storage locations, etc.

SUMMARY

According to an example, an online storage system stores files or more generally data. The storage system receives files to be stored in a database according to a placement policy. The storage system retrieves policy constraints described in the placement policy. The placement policy may include a plurality of copies to be stored for the file, wherein each copy has one or more policy constraints that constrain selection of storage locations. The storage system retrieves storage location constraints and storage location attributes of each storage location. The storage system determines a query for the first copy comprising the policy constraints for the first copy and the storage location constraints. The storage system extracts file attributes from the file. The file attributes may be file characteristics and/or other characteristics associated with the file. The storage system determines an attribute set for each storage location including the file attributes and the storage location attributes. The storage system identifies a set of candidate storage locations for storage of the first copy by evaluating the attribute sets against the query. The storage system selects a candidate storage location from the first set and provides the first copy of the file to the selected candidate storage location for storage.

In one or more embodiments, the storage system obtains health metrics for each of the storage location. The health metrics may be detected by the storage system through interactions with the storage locations, i.e., storing files, accessing files, etc. The health metrics of a storage location may be used to determine whether the storage location is eligible for selection consideration.

In one or more embodiments, the storage system ranks the candidate storage locations according to one or more ranking metrics. The ranking metrics provide some preference in the selection process. The candidate storage locations all have attribute sets that satisfy the query, yet the ranking preferentially orders the candidate storage locations.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** illustrates a networking environment for an online storage system, in accordance with one or more embodiments.

FIG. **2** illustrates an exemplary architecture of the online storage system, in accordance with one or more embodiments.

FIG. **3** illustrates a flow of files when stored by the online storage system, in accordance with one or more embodiments.

FIG. **4** illustrates a process flowchart for storing a file according to a placement policy, in accordance with one or more embodiments.

The figures depict various embodiments of the present invention for purposes of illustration only. One skilled in the art will readily recognize from the following discussion that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles of the invention described herein.

DETAILED DESCRIPTION

System Environment

FIG. **1** illustrates networking environment for an online storage system, in accordance with one or more embodiments. FIG. **1** includes a client device **110**, an online system **120**, a database **130**, and a network **140**. The online system **120** includes a storage system **150** that stores data or files (e.g., data objects, blobs, documents, etc.) obtained by the online system **120**. In some embodiments, the online system **120** is a transport service system that connects riders and drivers for transport service transactions. The online system **120**, in these embodiments, may also include a ride management system **160** that manages one or more aspects of the transport service transactions. For clarity, only one client device **110** is shown in FIG. **1**, but in reality, multiple client devices **110** may communicate with any component over the network **140**. Alternate embodiments of the system environment **100** can have any number online systems **120** and databases **130**. The functions performed by the various entities of FIG. **1** may also vary in different embodiments.

Users interact with the online system **120** through the client device **110**. The client device **110** can be personal or mobile computing devices, such as smartphones, tablets, or notebook computers. The client device **110** may interact with the online system **120** through client applications configured to interact with the online system **120**. Interactions include at least storing data or files to the online system **120**, providing instructions regarding storage of the files by the online system **120**, viewing files stored by the online system **120**, accessing files stored by the online system **120**, other interactions for file or data management, or any combination thereof. Files that are provided to the online system **120** may include text files, spreadsheets, photos, videos, data objects, other file types, or any combination thereof.

In embodiments of the online system **120** as a transport service system, users and drivers may interact with the client applications of the client devices **110** to request and access information about rides arranged. The client applications can present information received from the transport service system on a user interface, such as a map of the geographic region, the estimated trip duration, and other information. Additionally, the client devices **110** may provide their location and other data to the transport service system. For example, a current location of a client device **110** may be designated by a user or driver or detected using a location sensor of the client device (e.g., a global positioning system (GPS) receiver) and provided to the transport service system as coordinates. Some or all of the information—i.e., as files—regarding the ridesharing transactions may be provided to the online system **120** for storage.

The online system **120** stores files, e.g., via the storage system **150**. The storage system **150** obtains files, e.g., from one or more client devices **110**, to be stored. The storage system **150** stores the files in one or more storage locations **170** of the database **130**. The storage system **150** may be distributed over a plurality of servers, wherein each server may have server attributes. Example server attributes include, a geographical region where the server is deployed, whether the server is in a staging state (i.e., a state where the server handling file storage and retrieval is undergoing tests) or a production state (i.e., a state where the server is live with runtime file storage). The storage system **150** may have one or more global constraints (e.g., in effect for all files stored by the storage system **150**). A global constraint constrains which storage locations are selected. For example, a global constraint may indicate read-write accessibility with the storage location. In one or more embodiments, the storage system **150** stores files according to a placement policy (e.g., selected from a plurality of placement policies). A placement policy includes one or more policy constraints that constrain selection of storage locations for storage of files. The placement policy may also include one or more placement criteria that dictate placement of the file. The placement criteria may include any combination of a number of copies to store for the file, whether storage of a copy is mandatory or optional, a date and/or time when to store the file, a minimum health metric for storage locations to be eligible for selection consideration (e.g., remaining capacity), one or more ranking metrics for use in ranking candidate storage locations, synchronous or asynchronous upload-wherein a synchronous upload stores the file in the selected storage location temporally proximate to the upload time, whereas an asynchronous upload stores the file as a background operation not necessarily temporally proximate to the upload time, etc.

The storage system **150** selects storage locations to store copies of files. The storage system **150** generates an attribute set for each eligible storage location. The attribute set includes at least file attributes and storage location attributes. The file attributes are characteristics of the file, wherein storage location attributes are properties of the storage location. The attribute set may include additional attributes, e.g., one or more attributes of the storage system **150**, one or more attributes of a server of the storage system **150** handling the upload request, other attributes described herein this disclosure, etc. The storage system **150** generates a query according to the placement policy. Each eligible storage location **170** may also include one or more storage location constraints which constrain what files may be stored at the particular storage location. The query generated is a combination of global constraints, storage location constraints, and policy constraints. The attribute for each storage location is evaluated against the query to determine candidate storage locations that are eligible for use in storing the file. The storage system **150** may further rank the candidate storage locations according to a ranking metric. The storage system **150** selects one or more of the candidate storage locations and stores a copy of the file to the selected candidate storage location. In one or more embodiments, the storage system **150** obtains health metrics for each storage location. Storage locations having a poor health metric may be excluded from selection consideration. The storage system **150** will be further described in FIGS. **2**-**4**.

Selection of storage locations using attribute sets evaluated against queries to determine candidate storage locations provides greater flexibility in a placement policy. Due to the placement policy being agnostic of storage location, i.e., comprising policy constraints, there is no restriction to referring directly to storage locations. This agnosticism proves useful in situations when various storage locations are added, removed, or changed in some manner as the placement policy need not be rewritten to accommodate these various changes. For example, if a particular storage location has crashed (is in an off state), the placement policy as described herein can continue selection consideration with the remaining storage locations. Traditional placement policies dependent on particular storage locations would need to outline contingencies for every possible scenario. This placement policy allows for scalability as well. As more storage locations are added to the database **130**, the placement policies being agnostic to storage location can consider the newly added storage locations with the current storage locations.

In some embodiments, a ride management system **160** manages rideshare transactions. In managing rideshare transactions, the ride management system **160** may implement various algorithms for connecting riders and drivers. Each trip (e.g., a transport service that is requested and/or completed) can be logged, e.g., recording a date of the trip, a time of the trip, route data for a route traveled, a rider identifier, a driver identifier, a calculated cost for the trip, payment received, discount codes used, any delays, any excess fees, any notes, ratings, other trip information, etc. The ride management system **160** may provide information for a trip all at once or as each piece of information is received or calculated.

The database **130** stores files. The database **130** receives file from the online system **120**, or more specifically the storage system **150**. The database **130** stores the data comprised in a file. The database **130** comprises a plurality of storage locations **170**. One or more external systems may operate one or more of the storage locations **170**. For example, a first external system manages three storage locations with a second external system managing another one storage location. In any case, the database **130** may be networked to the online system **120** or directly communicative with the online system **120**.

The storage locations **170** are selected by the storage system **150** to store copies of files. In some embodiments, a storage location **170** is a physical disk storage that comprises various components and subcomponents relating to physical disk storage. In other embodiments, a storage location **170** is an online storage, e.g., an online cloud storage. Storage locations have one or more storage location attributes and may also have one or more storage location constraints.

The storage location attributes help to define the storage location. For example, a storage location attribute defines the technology of the storage location (s3, hdfs, glacier, etc.). As another example, a storage location attributes indicates accessibility of files stored, e.g., read only or read-write. Other storage location attributes include a geographical region of the storage location **170**, an accessibility speed of accessing stored files, a remuneration for storage of a file, etc.

Storage location constraints constrain what files may be stored at the particular storage location. For example, a storage location constraint constrains that files need to be greater than a size of 50 megabytes (mb) to be stored in a particular storage location. Other embodiments of storage location constraints may include any type of restrictions against various attributes of files, e.g., a storage location constraint excludes storage of files containing sensitive information. Another embodiment of storage location constraint constrains context of the uploaded file. In a first example, a storage location constraint permits only a specific group of one or more client devices that may store files at the respective storage location. In another example, a storage location constrains based on a geographical location of a server of the storage system **150** that receives the request to upload a file (e.g., only United States based storage systems **150** can upload to a particular storage location). The storage location attributes and the storage location constraints may be provided to the storage system **150** for use in selection consideration.

The various components of the system environment **100** communicate via one or more network interfaces to communicate over the network **140**. The network **140** comprises any combination of local area and wide area networks employing wired or wireless communication links. In some embodiments, all or some of the communication on the network **140** may be encrypted. For example, data encryption may be implemented in situations where the database **130** is located on a third-party online system separate from the online system **120**.

Storage System Architecture

FIG. **2** illustrates an exemplary architecture of the online storage system **150**, in accordance with one or more embodiments. The storage system **150** stores files received by the online system **120**. The storage system **150** selects one or more storage locations **170** and stores files at the selected storage locations. The storage system **150** has, among other components, a client interface module **210**, query composition module **220**, an attribute composition module **230**, a query evaluation module **240**, a location selection module **250**, a metric detection module **260**, at least one directory **270**A, a placement policy library **280**, a health metrics cache **285**, location configuration information **290**, and a query evaluation cache **295**. In other embodiments, the storage system **150** has additional or fewer components than those listed herein; moreover, the functions and operations of the various modules may also be interchanged amongst the modules. In embodiments not illustrated, the storage system **150** has a file retrieval module that retrieves files from the storage locations **170**, e.g., in response to requests for the file data.

The client interface module **210** moderates communication between the storage **150** and one or more client devices **110**. The client interface module **210** receives files and responds to requests from the client device **110**. The files are typically provided by the client device **110** for storage by the storage system **150**. Files may be any computer-readable file of any various file type. The file may be named, comprise file data, and other metadata properties. Metadata properties include size, file type, etc. In response to a client device **110** requesting access to files, the client interface module **210** distributes requested information to the client device **110**. In some instances, the information is presented in a GUI to the client device **110**.

The query composition module **220** generates one or more queries for use in selection of storage locations for use in storage of a file. The query composition module **220** generates a query with one or more global constraints, one or more storage location constraints, and one or more policy constraints, or any combination thereof. In some embodiments, the query generated is evaluated according to Boolean logic, wherein operators are used to combine the constraints in the query. The basic operators of Boolean logic include "AND", "OR", and "NOT" which are used to combine constraints in the query and determine how the constraints are satisfied during query evaluation. The "AND" operator combines two expressions such that both expressions need to be satisfied in order to resolve as true. The "OR" operator combines two expressions such that at least one of the two expressions need to be satisfied to resolve as true. The "NOT" operator inverses the resolution of an expression, e.g., from "false" to "true" and vice versa. Other more complicated operators may also be implemented in query composition. Other logics that may be implemented include first-order logic and other higher-order logics. Query languages that may be used include Structured Query Language (SQL), Object Query Language (OQL), Embedded SQL, Prolog, Datalog, other declarative logic programming languages, etc.

In one or more embodiments, a placement policy in effect indicates some number of copies of the file to be stored. Each copy specified by the placement policy may comprise individualized policy constraints used for selection of a storage location for storage of that copy. The query composition module **220** generates a query for each copy of the file to be stored. Accordingly, the storage system **150** selects a candidate storage location for each copy from the evaluation of the query for that copy.

The attribute composition module **230** generates an attribute set for each storage location eligible for selection consideration. The attribute composition module **230** generates an attribute set including one or more file attributes and one or more storage location attributes. The file attributes includes characteristics of the file. Example file attributes include a size of the file, a file type of the file, an uploading client device **110** that provided the file for storage, a geographical region where an uploading client device **110** provided the file for storage, a time of receipt of the file for storage, a sensitivity of data in the file, another known property of the file, etc. The file attributes are obtained from the file provided for storage. As mentioned above, storage location attributes help to define the storage location, e.g., technology of the storage location, a geographical region where the storage location is located, an accessibility of the storage location. The storage location attributes may be obtained from the storage locations **170** or may be recorded locally under location configuration information **290**.

In some embodiments, eligible storage locations are storage locations that have one or more health metrics above one or more thresholds. Eligible storage locations may be some or all of the storage locations **170** in the database **130**. For example, a success rate of storing a file above 95% is the determining factor as to whether a storage location is eligible for consideration. In an example with two health metrics, both health metrics of a storage location need to satisfy respective thresholds to be deemed eligible for selection consideration. In particular, this can be exampled with a storage location needing to have a success rate above 95% and a capacity rate below 50% in order to be deemed eligible. Accordingly, in embodiments with health metrics measured discretely (e.g., a storage location has an on state and an off state), eligibility may be achieved by having the appropriate discrete measure (e.g., the storage location needs to be in the on state to be eligible). In other embodiments, health metrics may also be considered as storage location attributes. As storage location attributes, the health metrics may be included in attribute sets generated for storage locations in selection consideration.

The query evaluation module **240** evaluates queries against attribute sets. The query evaluation module **240** evaluates a query against an attribute set according to the query logic. Accordingly, if the attribute set satisfies the query, the attribute set is evaluated to be true. In an example with Boolean logic, an attribute set satisfies a query with a number of constraints conjoined with the "AND" operator, if the attribute set includes at least one attribute that satisfies each of the constraints in the query. The attribute sets that satisfy a query are provided to the location selection module **250**. If N copies of a file are to be stored (as indicated by a placement policy) and considering M available storage locations, the query evaluation module **240** evaluates N×M evaluations. In one or more embodiments, the query evaluation module **240** caches one or more query evaluations in the query evaluation cache **295**. Cached query evaluations may be retrieved and used in evaluating subsequent queries.

The location selection module **250** selects a storage location and stores a copy of the file at the selected storage location. The location selection module **250** receives the query evaluations from the query evaluation module **240**. For a query, the location selection module **250** designates storage locations with attribute sets that satisfy the query as candidate storage locations. The location selection module **250** selects one of the candidate storage locations. The location selection module **250** stores a copy of the file to the selected candidate storage location. In some embodiments, the location selection module **250** may perform other processes prior to storage of the copy of the file, e.g., name hashing, file encryption, setting accessibility settings (e.g., password, particular client devices **110** that are allowed to access, etc.).

In one or more embodiments, the location selection module **250** refers to one or more ranking metrics in a placement policy for use in selecting the candidate storage location. For example, the ranking metric specifies ranking the candidate storage locations according to remuneration for storing a file. The location selection module **250** selects from the ranking the candidate storage location with the lowest remuneration for storing a file. In another example, the ranking metric specifies ranking the candidate storage locations according to proximity to the geographical region of the uploading client device **110**. The location selection module **250** may select from the ranking the candidate storage location in closest proximity to the geographical region of the uploading client device **110**. In additional embodiments, multiple ranking metrics are considered in tandem, wherein the location selection module **250** can rank according to a function of the multiple ranking metrics. For example, the function weights the influence of each ranking metric to generate an overall rank score for each candidate storage location.

The metric detection module **260** detects health metrics for one or more of the storage locations **170**. The metric detection module **260** may send requests (e.g., periodically) to each storage location **170** to provide various health metrics. For example, the metric detection module **260** may obtain a capacity of a storage location **170** as a health metric for the storage location **170**. The metric detection module **260** may also detect health metrics as files are provided for storage to the storage locations **170**. For example, the metric detection module **260** records a success or failure rate in storing files at a particular storage location **170** as one health metric. Another health metric may be retrieval success or failure rate. Other health metrics may be measured as relative latency in individual operations with the storage location, e.g., storage, retrieval, editing, etc.

The directory **270**A is used by the storage system **150** to organize files stored by the storage system **150**. The directory **270**A has configured policies **275**A put in place dictating operation of the directory **270**A. The configured policies **275**A includes a placement policy, e.g., selected from various placement policies stored under the placement policy library **280**. Each placement policy includes its own policy constraints. Other configured policies may include an indexing policy that dictates how files stored under the directory **270**A are indexed, an encryption policy that dictates how file data is encrypted, a retention policy that dictates circumstances in which files are deleted (e.g., files last accessed more than a year ago are deleted), etc. In additional embodiments, the storage system **150** comprises a plurality of directories **270** that may be arranged in a nested hierarchy. In embodiments with a nested hierarchy of directories, a child directory (nested under a parent directory) may inherit its parent directory's configured policies absent configured policies in place for the child directory.

FIG. **3** illustrates a flow **300** of a file during storage by the online storage system **150**, in accordance with one or more embodiments. In this illustration, there is one uploading client device **110**. The storage system **150** is a component of the online system **120**. In addition, there are two storage locations **170**A and **170**B, embodiments of storage locations **170**.

The uploading client device **110** provides a file to the storage system **150** for storage. The file **310** may be provided in a batch of other files. The file **310** has one or more file attributes that describe characteristics of the file. The storage system **150** receives the file **310**. The uploading client device **110** may specify a directory to store the file under. Accordingly, the storage system **150** retrieves configured policies for the specified directory.

The storage system **150** generates an attribute set for each eligible storage location, namely attribute set **350**A for storage location **170**A and attribute set **350**B for storage location **170**B. The attribute set includes one or more file attributes, one or more storage location attributes **315**, or any combination thereof. As described above, storage locations may be determined to be eligible with one or more health metrics above certain thresholds.

The storage system **150** generates a query **360** for one copy of the file. The query **360** generated includes one or more global constraints **320**, one or more storage location constraints **330**, one or more policy constraints **340**, or any combination thereof. The policy constraints **340** are retrieved from the placement policy in effect, e.g., set in place for the specified directory.

The storage system **150** evaluates each attribute set **350**A and **350**B against the query **360**. Storage locations with attribute sets evaluated to satisfy the query **360** are deemed to be candidate storage locations. In the example shown in FIG. **3**, the attribute set **350**A does not satisfy the query **360** and the query evaluation renders "false", while the attribute set **350**B does satisfy the query **360** such that the query evaluation is "true". The storage location **170**B is deemed a candidate storage location **370**. In this example with only storage location **170**B as the candidate storage location **370**, the storage system **150** provides a copy **380** of the file **310** to the storage location **170**B for storage. In some instances, no eligible storage location has an attribute set that satisfies the query **360** for a mandatory copy. These instances may arise if a placement policy is overly strict or if eligible storage locations are limited. In these instances, an error message may be sent back to the uploading client **110** indicating failure to effectively place a mandatory copy according to the placement policy. The error message may indicate which constraints in the query **360** were unsatisfied.

Placement Selection and Storing Files

FIG. **4** illustrates a process flowchart **400** for storing a file according to a placement policy, in accordance with one or more embodiments. The following steps are in the perspective of the storage system **150**, but can be performed by the various modules in the storage system **150** shown in FIG. **2**. In other embodiments, the storage system **150** may perform additional or fewer operations that those listed herein.

The storage system **150** receives **410**, from a client device **110**, a file to be stored in a database **130** according to a placement policy. The placement policy may be selected from a library of placement policies. Each placement policy comprises a plurality of policy constraints that dictate how many copies of the file are stored and aid in selection of the storage location to store each of the copies. The database **130** selects from a plurality of storage locations **170** for storing files, as described in FIG. **1**. Each file comprises various characteristics, e.g., a name of the file, file data, various properties of the file, a file type, etc. Moreover, other metadata properties may be associated with the file, such as, the uploading client, the geographical region of the uploading client, etc.

The storage system **150** retrieves **420** one or more policy constraints for a first copy of the file described in the placement policy. The placement policy may include a plurality of copies to be stored for the file, wherein each copy has one or more policy constraints that constrain selection of storage locations. Example policy constraints include constraining a particular geographical region, constraining a particular storage technology, constraining for a particular accessibility, etc.

The storage system **150** retrieves **430** one or more storage location constraints and one or more storage location attributes of each storage location of the plurality of storage locations. Each storage location has a set of storage location attributes that define the storage location. Storage location attributes include an accessibility (e.g., read only or read-write), a storage technology, a remuneration cost, etc. Storage location constraints may constrain what files may be stored in the storage location. Example storage location constraints include a minimum file size, a particular file type, etc. In some embodiments, some or all of the storage locations are absent storage location constraints.

The storage system **150** determines **440** a first query for the first copy comprising the policy constraints for the first copy and the storage location constraints. The first query is combines the various constraints. In some embodiments, the first query is a Boolean query wherein the constraints are combined with the "AND" operator. In embodiments with additional copies to be stored as defined in the placement policy, the storage system **150** determines a distinct query for each copy including policy constraints specified for that copy.

The storage system **150** extracts **450** one or more file attributes from the file. The file attributes may be file characteristics and/or other characteristics associated with the file.

The storage system **150** determines **460** an attribute set for each storage location including the file attributes and the storage location attributes. The attribute sets are a list of the file attributes and the storage location attributes. In some embodiments, an attribute set is generated for eligible storage locations. Eligibility in selection consideration may be determined according to one or more health metrics for the storage locations. The eligibility for a storage location may be evaluated based on whether the health metrics for the storage location are above a certain threshold.

The storage system **150** identifies **470** a first set of candidate storage locations for storage of the first copy by evaluating the attribute sets against the first query. The storage system **150** evaluates an attribute set against a query by evaluating whether the list of attributes in the attribute set satisfies the query. In an example of a Boolean query with all constraints combined with the "AND" operator, the storage system **150** evaluates whether each constraint is satisfied by at least one attribute of the attribute set. For example, if a policy constraint in the first query is "loc:tech:s3" constraining storage technology, then a storage location attribute of "loc:tech:s3" describing the storage location's storage technology satisfies that policy constraint. In examples with other operators, the storage system **150** evaluates whether the attribute set sufficiently evaluates "true" for the query. In embodiments where the placement policy includes additional copies, the storage system **150** evaluates the attribute set for each storage location against the query for each copy.

The storage system **150** selects **480** a first candidate storage location from the first set. The storage system **150** may rank the first set of candidate storage locations according to one or more ranking metrics. From the ranking, the storage system **150** selects the highest ranked candidate storage location for use in storing the first copy of the file.

The storage system **150** provides **490** the first copy of the file to the first candidate storage location for storage. The storage system **150**, upon storing the first copy, can detect one or more health metrics of the storage location. For example, depending on whether the storage of the first copy was successful or not, the storage system **150** may detect an average success rate in storing files at the storage location. In other embodiments, the storage system **150** may also obtain health metrics detected by the storage location. For example, the storage location provides a capacity of the storage location as a health metric to the storage system **150**.

Additional Configuration Information

The foregoing description of the embodiments of the disclosure has been presented for the purpose of illustration; it is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Persons skilled in the relevant art can appreciate that many modifications and variations are possible in light of the above disclosure.

Some portions of this description describe the embodiments of the disclosure in terms of algorithms and symbolic representations of operations on information. These algorithmic descriptions and representations are commonly used by those skilled in the data processing arts to convey the substance of their work effectively to others skilled in the art. These operations, while described functionally, computationally, or logically, are understood to be implemented by computer programs or equivalent electrical circuits, microcode, or the like. Furthermore, it has also proven convenient at times, to refer to these arrangements of operations as modules, without loss of generality. The described operations and their associated modules may be embodied in software, firmware, hardware, or any combinations thereof.

Any of the steps, operations, or processes described herein may be performed or implemented with one or more hardware or software modules, alone or in combination with other devices. In one embodiment, a software module is implemented with a computer program product comprising a computer-readable medium containing computer program code, which can be executed by a computer processor for performing any or all of the steps, operations, or processes described.

Embodiments of the disclosure may also relate to an apparatus for performing the operations herein. This apparatus may be specially constructed for the required purposes, and/or it may comprise a general-purpose computing device selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a non-transitory, tangible computer readable storage medium, or any type of media suitable for storing electronic instructions, which may be coupled to a computer system bus. Furthermore, any computing systems referred to in the specification may include a single processor or may be architectures employing multiple processor designs for increased computing capability.

Embodiments of the disclosure may also relate to a product that is produced by a computing process described herein. Such a product may comprise information resulting from a computing process, where the information is stored on a non-transitory, tangible computer readable storage medium and may include any embodiment of a computer program product or other data combination described herein.

Finally, the language used in the specification has been principally selected for readability and instructional purposes, and it may not have been selected to delineate or circumscribe the inventive subject matter. It is therefore intended that the scope of the disclosure be limited not by this detailed description, but rather by any claims that issue on an application based hereon. Accordingly, the disclosure of the embodiments is intended to be illustrative, but not limiting, of the scope of the disclosure, which is set forth in the following claims.

What is claimed is:

1. A method comprising:

receiving, from a client device, a request for storing a file in a directory;

retrieving, from a placement policy library, a placement policy associated with the directory, the placement policy specifying:

a plurality of copies to be stored, and

a set of policy constraints for each of the plurality of copies, the set of policy constraints including

(1) an attribute set of storage location to which a corresponding copy is to be stored; and

(2) required upload mode selected from synchronous or asynchronous, wherein the policy constraints for the plurality of copies include a first set of policy constraints associated with a first copy of the plurality of copies of the file, and a second set of policy constraints associated with a second copy of the plurality of copies of the file, and wherein the first or second set of policy constraints specify (1) an attribute set of storage to which the respective first or second copy of file is to be stored, and (2) whether the respective first or second copy of file is uploaded using a synchronous or asynchronous upload, and wherein a synchronous upload stores the file temporarily proximate to an update time, and an asynchronous upload stores the file as a background operation;

determining a current attribute set for each of a plurality of storage locations;

generating a data structure storing current attribute sets of the plurality of storage locations;

determining a first query in a structured query language for the first copy of the file, the first query comprising the first set of policy constraints for the first copy;

executing the first query against the data structure storing the current attribute sets for storage locations to identify a first set of one or more candidate storage locations for storage of the first copy based on evaluation of the current attribute sets of the plurality of storage locations against the first query;

selecting a first candidate storage location from the first set of one or more candidate storage locations for storage of the first copy;

providing the first copy of the file to the first candidate storage location for storage;

determining a second query in a structured query language for the second copy comprising the second set of policy constraints for the second copy;

executing the second query against the data structure storing the current attribute sets for storage locations to identify a second set of one or more candidate storage locations for storage of the second copy based on evaluation of the current attribute sets of the plurality of storage locations against the second query;

selecting a second candidate storage location from the second set of one or more candidate storage locations for storage of the second copy; and

providing the second copy of the file to the second candidate storage location for storage.

2. The method of claim **1**, wherein identifying the first set of one or more candidate storage locations for storage of the first copy by executing the first query against the data structure storing the current attribute sets comprises:

for each storage location of the plurality of storage locations, determining whether the current attribute set satisfies the first query,

wherein the first set of one or more candidate storage locations include storage locations whose current attribute set satisfy the first query.

3. The method of claim **1**, further comprising:

obtaining one or more health metrics for each storage location of the plurality of storage locations, and

selecting the first candidate storage location or the second candidate storage location further based on the health metrics of the plurality of storage locations.

4. The method of claim **3**, wherein a storage location is excluded from the plurality of storage locations if the first health metric of the storage location is below a threshold.

5. The method of claim **3**, wherein the one or more health metrics of a storage location comprise at least one of: a success rate, a failure rate, a capacity usage rate, or a relative latency in individual operations with the storage location.

6. The method of claim **1**, further comprising:

receiving an input to adjust the placement policy; and

responsive to the input, adjusting one or more policy constraints described in the placement policy.

7. The method of claim **1**, wherein the file is associated with one or more file attributes, and the plurality of policy constraints are associated with the one or more file attributes.

8. The method of claim **7**, wherein the one or more file attributes comprise at least one of: file size, file type, or geographical origin of the uploading client device.

9. A non-transitory computer-readable storage medium storing instructions that, when executed by one or more processors, cause the one or more processor to perform operations comprising:

receiving, from a client device, a request for storing a file to be storedin a directory;

retrieving, from a placement policy library, a placement policy associated with the directory, the placement policy specifying:

a plurality of copies to be stored, and

a set of policy constraints for each of the plurality of copies, the set of policy constraints including

(1) an attribute set of storage location to which a corresponding copy is to be stored; and

(2) required upload mode selected from synchronous or asynchronous, wherein the policy constraints for the plurality of copies include a first set of policy constraints associated with a first copy of the plurality of copies of the file, and a second set of policy constraints associated with a second copy of the plurality of copies of the file, and wherein the first or second set of policy constraints specify (1) an attribute set of storage to which the respective first or second copy of file is to be stored, and (2) whether the respective first or second copy of file is uploaded using a synchronous or asynchronous upload, and wherein a synchronous upload stores the file temporarily proximate to an update time, and an asynchronous upload stores the file as a background operation;

determining a current attribute set for each of a plurality of storage locations;

generating a data structure storing current attribute sets of the plurality of storage locations;

determining a first query in a structured query language for the first copy of the file, the first query comprising the first set of policy constraints for the first copy;

executing the first query against the data structure storing the current attribute sets for storage locations to identify a first set of one or more candidate storage locations for storage of the first copy based on evaluation of the current attribute sets of the plurality of storage locations against the first query;

selecting a first candidate storage location from the first set of one or more candidate storage locations for storage of the first copy;

providing the first copy of the file to the first candidate storage location for storage;

determining a second query in a structured query language for the second copy comprising the second set of policy constraints for the second copy;

executing the second query against the data structure storing the current attribute sets for storage locations to identify a second set of one or more candidate storage locations for storage of the second copy based on evaluation of the current attribute sets of the plurality of storage locations against the second query;

selecting a second candidate storage location from the second set of one or more candidate storage locations for storage of the second copy; and

providing the second copy of the file to the second candidate storage location for storage.

10. The non-transitory computer-readable storage medium of claim 9, wherein identifying the first set of one or more candidate storage locations for storage of the first copy by executing the first query against the data structure storing the current attribute sets comprises:

for each storage location of the plurality of storage locations, determining whether the current attribute set satisfies the first query,

wherein the first set of one or more candidate storage locations include storage locations whose current attribute set satisfy the first query.

11. The non-transitory computer-readable storage medium of claim 9, wherein the instructions further cause the one or more processors to:

obtaining one or more health metrics for each storage location of the plurality of storage locations, and

selecting the first candidate storage location or the second candidate storage location further based on the health metrics of the plurality of storage locations.

12. The non-transitory computer-readable storage medium of claim 11, wherein a storage location is excluded from the plurality of storage locations if the first health metric of the storage location is below a threshold.

13. The non-transitory computer-readable storage medium of claim 11, wherein the one or more health metrics of a storage location comprise at least one of: a success rate, a failure rate, a capacity usage rate, or a relative latency in individual operations with the storage location.

14. The non-transitory computer-readable storage medium of claim 9, wherein the instructions further cause the one or more processors to:

receive an input to adjust the placement policy; and

responsive to the input, adjust one or more policy constraints described in the placement policy.

15. The non-transitory computer-readable storage medium of claim 9, wherein the file is associated with one or more file attributes, and the plurality of policy constraints are associated with the one or more file attributes.

16. The non-transitory computer-readable storage medium of claim 15, wherein the one or more file attributes comprise at least one of: file size, file type, or geographical origin of the uploading client device.

17. A computer system comprising:

one or more processors; and

a non-transitory computer-readable storage medium storing instructions that, when executed by the one or more processors, cause the one or more processors to perform operations comprising:

receiving, from a client device, a request for storing a file in a directory;

retrieving, from a placement policy library, a placement policy associated with the directory, the placement policy specifying:

a plurality of copies to be stored, and

a set of policy constraints for each of the plurality of copies, the set of policy constraints including

(1) an attribute set of storage location to which a corresponding copy is to be stored; and

(2) required upload mode selected from synchronous or asynchronous, wherein the policy constraints for the plurality of copies include a first set of policy constraints associated with a first copy of the plurality of copies of the file, and a second set of policy constraints associated with a second copy of the plurality of copies of the file, and wherein the first or second set of policy constraints specify (1) an attribute set of storage to which the respective first or second copy of file is to be stored, and (2) whether the respective first or second copy of file is uploaded using a synchronous or asynchronous upload, and wherein a synchronous upload stores the

file temporarily proximate to an update time, and an asynchronous upload stores the file as a background operation;

determining a current attribute set for each of a plurality of storage locations;

generating a data structure storing current attribute sets of the plurality of storage locations;

determining a first query in a structured query language for the first copy of the file, the first query comprising the first set of policy constraints for the first copy;

executing the first query against the data structure storing the current attribute sets for storage locations to identify a first set of one or more candidate storage locations for storage of the first copy based on evaluation of the current attribute sets of the plurality of storage locations against the first query;

selecting a first candidate storage location from the first set of one or more candidate storage locations for storage of the first copy;

providing the first copy of the file to the first candidate storage location for storage;

determining a second query in a structured query language for the second copy comprising the second set of policy constraints for the second copy;

executing the second query against the data structure storing the current attribute sets for storage locations to identify a second set of one or more candidate storage locations for storage of the second copy based on the evaluation of the current attribute sets of the plurality of storage locations against the second query;

selecting a second candidate storage location from the second set of one or more candidate storage locations for storage of the second copy; and

providing the second copy of the file to the second candidate storage location for storage.

18. The computer system of claim **17**, wherein identifying the first set of one or more candidate storage locations for storage of the first copy by executing the first query against the data structure storing the current attribute sets comprises:

for each storage location of the plurality of storage locations, determining whether the current attribute set satisfies the first query,

wherein the first set of one or more candidate storage locations include storage locations whose current attribute set satisfy the first query.

19. The computer system of claim **17**, wherein the one or more processors are further caused to:

obtain one or more health metrics for each storage location of the plurality of storage locations, and

select the first candidate storage location or the second candidate storage location further based on the health metrics of the plurality of storage locations.

20. The computer system of claim **19**, wherein the one or more health metrics of a storage location comprise at least one of: a success rate, a failure rate, a capacity usage rate, or a relative latency in individual operations with the storage location.

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